Voxel Camera with Laser Tracking

Milestone M2 — Requirements and Gate 1

Team: Voxel Camera with Laser Tracking
Milestone: M2
Date: 12 November 2026

Gate 1. The team has read and approved the requirements in this report on 12 November 2026.

[L] = lecture slides “Distributed Real-Time Systems”, winter term 2026, 308 pages; “p.” is the PDF page number. Lecture pages justify every requirement with a time; the numeric values are the project’s own choices unless stated otherwise.

1. The system in a few sentences

A lab demo that shows multi-camera 3D tracking. Three Raspberry Pi cameras stand at right angles around a cube; a ball swings on a string through the cube. The system computes the 3D position of the ball and steers a laser through a galvo mirror pair so that the spot stays on the ball. The cameras also see the spot, so the system measures its own aiming error. Built and operated by a team of 11 students, presented once to other students.

Field of application: a distributed real-time system with embedded sensing and actuation — five Raspberry Pi 3 nodes (three camera nodes, one control node, one galvo node) on one wired Ethernet switch with a shared software PTP clock, built as a laboratory demonstrator.

The physical loop runs from the swinging ball and the laser spot, through the three cameras, the network and the control node (3D position and prediction), to the galvo mirrors and the laser, and back to the spot on the ball, which the cameras observe again. The cameras therefore see the effect of the actuator, so the system measures its own aiming error.

1.1. Terms

Term	Meaning
cube	the tracking volume and its coordinate system: right-handed, in mm; origin at the floor corner marked “0” on the frame, x and y along the floor edges marked “X” and “Y”, z up. All positions in requirements and tests are given in it.
ball	the tracked object, ≈ 5 cm diameter, on a ≈ 1 m string
spot	the laser spot as seen by the cameras
camera node / control node / galvo node	the three node types (ADR-VLT-001)
observation	one camera’s detection result for one frame: ball and spot image positions plus exposure time
ball state	fused 3D ball position (and velocity) at a time
target	the cube point the galvo node shall aim at, with its validity time and laser flag
aiming error	distance in mm from the fused ball centre to the laser beam line (galvo origin ⟶ spot in 3D); unmeasurable when the spot cannot be placed in 3D (ADR-VLT-005)
hit	a fusion slot in which, in at least one camera that sees both, the spot lies inside the ball’s image disc (ADR-VLT-005)
run	from release (ball displaced > 100 mm starts moving) until the amplitude falls below 50 mm; aborted when the safe state ends it (ADR-VLT-006)
hit rate	time with a hit ÷ run duration; gaps count as misses (ADR-VLT-006)
fusion slot	one period of the fixed-rate fusion, aligned to the shared clock (ADR-VLT-009)
arming epoch	counter incremented at every arming; the galvo node needs a newer one after a safety fault (ADR-VLT-007)
rest position	the ball centre while the ball hangs still
amplitude	the ball’s maximum horizontal distance from its rest position over the last swing
armed	the state in which the system may switch the laser on
safe state	laser off; galvo node ignores targets until re-armed
degraded	tracking with exactly two camera nodes delivering the ball
camera calibration / galvo calibration	stored results of the automatic calibrations (ADR-VLT-012)
reference data set	recorded, annotated frames of all cameras used by the automated tests (NF-VLT-21)
monitoring laptop	observe-only sink for monitoring data (ADR-VLT-003)
kill switch	physical switch cutting the laser supply independent of software
sponsor	the product-owner role of the project: sets the goal of the demo and decided the questions marked “sponsor”; for the team its decisions are given
setup documentation	the team’s record of the built setup: hardware list, wiring, laser-safety procedure and responsible role; delivered with the demo

1.2. How to read the requirements

Every requirement has an ID (F- functional, NF- non-functional), a reason with its lecture basis [L] and a falsifiable acceptance criterion. Every time carries a unit. IDs are never renumbered; a dropped requirement keeps its ID. Requirements reference

decisions (ADR-VLT-n, Section 5) and open questions (OQ-n, Section 6) by ID. The real-time transactions T1 … T7 and their deadline classes are defined in NF-VLT-26.

1.3. Requirements at a glance

Topic	Requirements
Demo goal and evaluation	NF-VLT-1, F-VLT-6, F-VLT-7
Sensing and 3D position	F-VLT-1 … F-VLT-4, F-VLT-12, F-VLT-27, NF-VLT-17, NF-VLT-21
Actuation	F-VLT-5, F-VLT-9, F-VLT-13, F-VLT-15, F-VLT-19, NF-VLT-16, NF-VLT-32
Time base and deadlines	NF-VLT-5, NF-VLT-18, NF-VLT-24, NF-VLT-26, NF-VLT-33, F-VLT-18, F-VLT-20, F-VLT-22
Laser safety	NF-VLT-2, NF-VLT-3, NF-VLT-25, NF-VLT-30, F-VLT-8, F-VLT-10, F-VLT-16, F-VLT-21, F-VLT-26, F-VLT-30, F-VLT-42
Fault tolerance and error detection	NF-VLT-23, NF-VLT-34, F-VLT-11, F-VLT-17, F-VLT-23, F-VLT-24, F-VLT-25
Modes, start-up and operation	F-VLT-28, F-VLT-31, F-VLT-33, F-VLT-35, F-VLT-41
Monitoring and diagnostics	F-VLT-29, F-VLT-32, F-VLT-34, F-VLT-37 … F-VLT-40, NF-VLT-6 … NF-VLT-9
Resources, load, security, testability and project standards	NF-VLT-10 … NF-VLT-15, NF-VLT-19, NF-VLT-20, NF-VLT-22, NF-VLT-27 … NF-VLT-29, NF-VLT-31
Dropped or deferred (IDs kept)	F-VLT-14, F-VLT-36, NF-VLT-4

2. Functional requirements

F-VLT-1 Each camera node detects the ball in every captured frame and sends the control node an observation. The observation contains the ball's image position and image radius (or "not seen") and the exposure time in the shared clock.
Reason. Fusion and prediction need the time of the measurement, not the arrival time. The hit decision (F-VLT-7) needs the ball's image disc. [L] p. 111–114 (observation, update delay).
Acceptance. On the reference data set (NF-VLT-21): the detected ball position lies within 3 px of the annotation in ≥ 95 % of the frames with an annotated ball, and the detected image radius lies within 2 px of the annotation in ≥ 95 % of them; frames without the ball report "not seen". The 3 px / 2 px / 95 % bounds also hold when evaluated only over frames with the spot on the ball. Automated test without hardware; the same test on a camera node gives identical results.

F-VLT-2 Each camera node detects the laser spot in every captured frame and reports its image position (or "not seen") in the same observation.
Reason. The effect of the actuator must be observed, not only commanded. [L] p. 206–207 (end-to-end protocol).
Acceptance. On the reference data set: the detected spot lies within 3 px of the annotation in ≥ 95 % of the frames with an annotated spot; frames recorded with the laser off report "not seen" in ≥ 99 %. Automated test without hardware; identical results on a camera node.

F-VLT-3 The control node computes the ball's 3D position in cube coordinates for one common instant from the observations of at least two camera nodes and the camera calibration. Observations taken at different times are first brought to that instant using their exposure times. New positions are produced at a fixed period aligned to the shared clock (ADR-VLT-009).
Reason. The 3D position is a real-time image of a moving entity: only observations that refer to the same instant may be combined. [L] p. 97–98 (timestamps across nodes), 113–116 (RT image, temporal accuracy), 185 (time-triggered).
Acceptance. On the reference data set: for the resting positions the computed position lies within 10 mm of the hand-measured position with 3 cameras and within 15 mm with any 2 cameras (automated, no hardware). Over a 10-min run the start times of the position updates deviate ≤ 1 ms from the configured period grid.

F-VLT-4 The control node aims at the position the ball will have when the spot is placed (compensating the loop delay), not at the last measured position. The prediction must not assume a pendulum motion (ADR-VLT-011).
Reason. At 1 m/s the ball moves ≈ 30 mm during a 30 ms loop, more than its radius of 25 mm. [L] p. 114 (update delay), 116–120 (temporal accuracy).

Acceptance. Replay of the recorded swings of the reference data set: the predicted position for the placement time deviates ≤ 10 mm (median) and ≤ 20 mm (95th percentile) from the recorded position at that time. Replay of a synthetic straight-line and circular motion within the operating envelope (NF-VLT-1) meets the same bounds.

F-VLT-5 The galvo node points the beam at each commanded cube point using the galvo calibration and places the spot at the time requested by the control node. Between two targets it keeps following the motion predicted by the control node. It never uses a target after the target's validity has expired (ADR-VLT-009).
Reason. A correct estimate has no value without a correct and timely physical aim; an expired target is an RT image that is no longer temporally accurate. [L] p. 116 (temporal accuracy), 146–148 (permanence, action delay).
Acceptance. Laser aimed at 9 known cube points: the spot lands within 5 mm of each. Replay of a target stream with known motion: between targets the output follows the predicted motion within 0.5 mm; the first output of each target is applied within 1 ms of its requested time; an expired target is never applied.

F-VLT-6 The control node computes the aiming error for every position update: the distance in mm from the fused ball centre to the laser beam line (galvo origin ⟶ spot in 3D). The spot in 3D is the near intersection of a camera's line of sight to the spot with the ball sphere (measured ball radius) when the spot lies inside the ball's image disc, or the triangulated spot when at least two cameras see it off the ball. Otherwise the aiming error is reported as unmeasurable, never as 0 (ADR-VLT-005).
Reason. The system must measure its own result to be self-checking; the metric is fixed by ADR-VLT-005. [L] p. 206–207 (end-to-end protocol).
Acceptance. On the reference data set: the computed aiming error lies within 5 mm of the error derived with the same geometry from the annotations and the hand-measured or fused 3D ball position; frames with a spot seen by one camera off the ball report unmeasurable.

F-VLT-7 The control node detects runs and reports their result. **Rest position:** part of the setup, measured once by hand. Whenever the ball hangs still (speed < 20 mm/s for ≥ 3 s) within 30 mm of the stored value, the control node updates it to the measured position. Without a known rest position no run starts. **Start (release):** the ball was held still (speed < 20 mm/s for ≥ 0.5 s) more than 100 mm from rest, then starts moving towards the rest position faster than 100 mm/s. Carrying the ball outward by hand doesn't start a run. **End:** the amplitude (largest horizontal distance from rest over the last swing period, ≈ 2 s) falls below 50 mm. **Hit:** in each position update, each camera's observation with the exposure time closest to the update time is used, if it is at most one frame period away. The update is a hit if, in at least one of these observations, ball and spot are both seen and the spot lies inside the ball's image disc (ADR-VLT-005). **Score:** hit rate = time with a hit ÷ run duration. Every update weighs the same. An update without valid ball state, without target or with the laser off is a miss. A run ended by the safe state is reported as aborted (ADR-VLT-006). The result reports duration, hit time, gap time and outcome. All thresholds are configurable; the values above are the defaults.
Reason. The demo goal (NF-VLT-1) needs a repeatable, unbiased evaluation that judges the measured effect, not the target. [L] p. 111–113 (observations), 206 (end-to-end protocol).
Acceptance. Replay of a recorded run of the reference data set, including holding the ball before release and gaps without valid position: start and end detected within one swing period (≈ 2 s) of the annotated moments; reported hit rate equals the hand count ± 1 %; gap time equals the injected gaps. Holding the ball at 150 mm for 5 s does not start a run. Carrying the ball outward by hand at 200 mm/s, with and without pauses of 1 s, does not start a run. A stored rest position 50 mm off: no run starts, and a diagnostic is shown.

F-VLT-8 The laser is off at start-up, during the restart of the control node or the galvo node, whenever the system is not armed and whenever no valid ball state is available. The restart of a single camera node is governed by F-VLT-11/17 (ADR-VLT-007). Exceptions: the galvo calibration scan (F-VLT-13) and the arming check (F-VLT-31) (ADR-VLT-012/013). Whenever the laser is on, F-VLT-42 applies.
Reason. Actuation is allowed only with valid prerequisites; every node starts in a safe ground state. [L] p. 79 (ground state), 217 (safety), 52–53 (hard deadline).
Acceptance. Power-up, control/galvo node restarts and covering the ball: a photodiode at the galvo output shows the laser off within 100 ms in every case, and no on-pulse occurs during start-up. A camera node restart with the other two tracking: behaviour per F-VLT-11.

F-VLT-9 The galvo node clamps every aim to the cube volume, regardless of the target it receives.
Reason. Received values must not bypass the local actuator limits. [L] p. 247 (fault-tolerant actuators).
Acceptance. Targets outside the cube (incl. NaN, ±∞, far out of range): the commanded mirror angles never leave the cube's angle envelope; NaN/∞ targets switch the laser off.

F-VLT-10 The galvo node switches the laser off when it has not received a fresh valid target for 100 ms, measured with its local clock.

Reason. Loss of the commanding node or the network must not leave the laser on (transaction T2, hard, NF-VLT-26). [L] p. 52–53 (deadlines), 163 (monitoring the message flow).
Acceptance. Stop the control node or unplug its cable while tracking: laser off ≤ 100 ms + 10 ms after the last target (photodiode). Invalid messages do not restart the timeout.

F-VLT-11 With exactly two camera nodes delivering the ball, tracking continues in degraded mode without re-arming. A camera "delivers" when its latest observation is valid (F-VLT-22/23) and sees the ball. The control node distinguishes and reports per camera (F-VLT-34): *absent* (no valid observation within the observation-age limit), *unsynced*, *present, ball not seen* and *present, ball seen*. If fewer than two cameras see the ball while at least two are present, the laser is off, and tracking resumes without re-arming if two see it again within 1 s. Otherwise, and whenever fewer than two cameras are present, the safe state is entered (ADR-VLT-007).
Reason. One camera fault is tolerated (k = 1, NF-VLT-23); the system needs to know which nodes are active to enter degraded mode. [L] p. 60 (fault hypothesis), 243 (membership, degraded mode).
Acceptance. Cover one camera, then stop one camera: tracking continues in both cases, mode = degraded naming that camera, and the reported camera status differs (ball not seen vs. absent). Cover the ball for two cameras for 0.5 s: laser off ≤ 100 ms, resumes without re-arming; for 2 s: safe state, re-arming needed. Stop two cameras: safe state.

F-VLT-12 Camera calibration is automatic (ADR-VLT-012). The reference is a set of markers fixed to the cube frame at positions measured once by hand. At every start each camera node calibrates itself without operator action. While running, a camera that has moved is detected, treated as absent (F-VLT-11) and named in a diagnostic. A camera that can't confirm its calibration at start stays absent.
Reason. Repeated setup tests need calibration in seconds; a moved camera would corrupt the fusion without any visible sign. [L] p. 162 (error detection), 240 (reintegration).
Acceptance. After an automatic calibration the F-VLT-3 acceptance passes, with a marker reprojection error ≤ 0.5 px. Calibration at start completes in ≤ 10 s per camera. Turning one camera by a few degrees during tracking: that camera is excluded within 1 s, tracking continues degraded; after a restart in the new position it calibrates itself again and F-VLT-3 passes.

F-VLT-13 Galvo calibration is an automatic scan (ADR-VLT-012). The galvo steps a grid of spots onto the cube's floor and walls, the cameras measure every spot seen by at least two of them, and a model of the galvo is fitted to the measurements. The control node starts the scan by itself when no valid galvo calibration exists or the arming check fails (F-VLT-31); there is no operator trigger (ADR-VLT-013). The laser is on only during the scan. Pressing the kill switch or any safety fault (F-VLT-21) aborts the scan with the laser off. Loss of the measurable spot (F-VLT-42) fails the scan; the laser is off before any further scan. After 2 consecutive failed scans (configurable), no further scan starts until the system is restarted, and a diagnostic names the reason. Commanded angles stay within the galvo limits of the cube (F-VLT-9). The result is stored and used until the next scan.
Reason. The galvo must be calibrated without manual work and without uncontrolled laser use. [L] p. 217 (safety), 254 (reaction matrix).
Acceptance. Start without a galvo calibration: the scan starts by itself, completes in ≤ 60 s and reports the model residual; afterwards F-VLT-5 passes. Pressing the kill switch mid-scan: scan aborted, previous calibration kept. Spot covered so that every scan fails: each scan fails by F-VLT-42 with the laser off before the next one; after 2 attempts no further scan, laser off, diagnostic shown. After restart the same calibration is loaded.

F-VLT-14 Deferred by the sponsor (2026-11-10, OQ-8) — was: optional redundant control node with takeover ≤ 1 s.
Reason. Conflicts with re-arming after the target timeout and needs a membership protocol and a second clock reference. [L] p. 231–234 (replication), 243.
Acceptance. —

F-VLT-15 The galvo node drives the galvo driver's X/Y inputs within the driver's input specification, through the converter board (PCM5102A I2S DAC, given hardware, ADR-VLT-015). From power-up, after a galvo node reset and whenever its output stops, the output holds the mirrors at the centre. If the output stream is interrupted (e.g. the output buffer runs empty), the galvo node switches the laser off within 100 ms. A broken link to the converter board isn't checked separately and can't be read back: it is detected only when it produces an effect that F-VLT-26 or F-VLT-42 detects; a fault without such an effect has no guaranteed detection time (ADR-VLT-015/018).
Reason. The actuator needs a defined output in the ground state; an interrupted output must not keep the laser on at an uncontrolled aim. [L] p. 79 (ground state), 202 (I/O error handling), 206 (end-to-end protocol).
Acceptance. Full-range X/Y sweep measured at the driver input stays within the driver's specified range and covers the cube's angle envelope. Power-cycling the galvo node keeps the driver input at the centre value throughout boot (scope trace); unplugging the I2S clock puts the mirrors at the centre. Starving the output stream (test profile): laser off within 100 ms (photodiode). Unplugging the I2S data line while armed, in a scene where this produces a persistent aiming error > 50 mm or loss of the measurable spot: laser off and alarm within 600 ms

from the origin stated in F-VLT-26/42, with the time of the electrical fault recorded separately. A fault without such an effect is recorded as outside the detection guarantee. (open: OQ-4)

F-VLT-16 If a camera sees the spot in a frame whose exposure started later than the actual laser-off time reported by the galvo node (F-VLT-25), plus the laser switch-off delay, plus one frame period, the system raises an alarm and enters the safe state (latched, F-VLT-21). Spots in frames exposed within that action-delay window after a switch-off don't raise the alarm.
Reason. The effect is monitored, not the target; a message may be acted on only after its action delay. [L] p. 146–148 (permanence, action delay), 206 (end-to-end protocol).
Acceptance. Inject a spot-like light (second pointer) while disarmed: alarm shown, safe state. Switch the laser off 20 × during tracking (kill switch, covering the ball): no alarm.

F-VLT-17 A restarted camera node rejoins the running system without restarting other nodes.
Reason. A recoverable camera outage must not stop the demo. [L] p. 79 (h-state, ground state), 240 (reintegration). Transaction T5, no deadline (NF-VLT-26).
Acceptance. Restart one camera node during tracking: it delivers observations again after its clock is synchronized; mode returns from degraded to normal.

F-VLT-18 Every node measures its clock offset to the shared clock and reports it. The monitoring display derives the worst pairwise offset (max − min over all synchronized nodes) and flags it when it exceeds π = 1 ms (NF-VLT-24).
Reason. The precision of the global time must be measured, not assumed. [L] p. 92 (precision), 100 (synchronization condition).
Acceptance. Values appear on the display for all 5 nodes, updated at least once per second, together with the worst pairwise offset. The independent cross-check of NF-VLT-5 (LED flash) is still required, because these are the nodes' own estimates.

F-VLT-19 If the galvo set contains its own pattern player, it is disabled or bypassed so that only the galvo node drives the mirrors. (The given set lists only galvo and driver, ADR-VLT-015; on the bench no pattern board is in the chain.)
Reason. No second, uncontrolled source may drive the actuator. [L] p. 247.
Acceptance. With the galvo node stopped, the mirrors don't move and no pattern is projected.

F-VLT-20 A node's observations or targets are used only while its clock is synchronized within 0.5 ms ($\pi/2$) of the clock reference (NF-VLT-24); until then the laser stays off.
Reason. Timestamps of unsynchronized nodes can't be compared. [L] p. 92, 97 (reasonableness), 100.
Acceptance. Start the system with clock synchronization disabled on one camera: its observations are ignored. Disabled on all cameras or on the galvo node: the laser stays off. Offset forced to 0.6 ms on one camera (test profile): its observations are ignored.

F-VLT-21 After a safety fault, the laser stays off until the control node arms again per F-VLT-35. Safety faults are: kill switch pressed (F-VLT-30), target timeout, galvo or control restart, sync loss leaving fewer than two usable cameras or on the galvo node, alarm (F-VLT-16/26, or F-VLT-42 while tracking; a converter link fault counts through its observed effect, ADR-VLT-015/018), interrupted galvo output stream (F-VLT-15), fewer than two cameras present, and ball not seen by two cameras for > 1 s. Re-arming happens only after the fault has cleared and ≥ 2 cameras have seen the ball continuously for 2 s (configurable) (ADR-VLT-007/013). The galvo node itself refuses to switch the laser on after a safety fault until it has seen a new arming, even if status reports between the nodes were lost. (ADR-VLT-013) An alarm, or 3 safety faults within 60 s (configurable), latches the safe state; the latch clears only when the kill switch is pressed and released. Loss of the measurable spot during a galvo scan or arming check is not an alarm: it fails that scan or check (F-VLT-13/31/42).
Reason. Safety-relevant shutdowns need a deliberate resume; without an arming button, the kill switch provides the deliberate act. [L] p. 217 (safety), 254 (reaction matrix).
Acceptance. For each listed fault: the laser stays off after the fault clears until the 2 s hold-off has passed. Suppress all galvo status reports and inject a target timeout: the galvo node still keeps the laser off for targets from before the fault. Inject an alarm (including F-VLT-42 while tracking), or 3 target timeouts within 60 s: the system stays safe after the hold-off and re-arms only after the kill switch was pressed and released. Spot loss during a scan or check: that scan or check fails, no latch.

F-VLT-22 Data used for fusion and aiming must meet explicit limits: observation age, difference of exposure times between cameras, prediction horizon and target validity. A new target based on expired observations is not fresh. Clock synchronization is checked throughout operation, including the age of the sync status. Invalid data is excluded, and the laser-off rules apply when no valid input remains. A correction of the shared clock must not extend the local target timeout (F-VLT-10). The limits are derived from the process: tolerated position error ÷ maximum

ball speed of the operating envelope (NF-VLT-1). Their values are chosen by the team and set in the architecture (OQ-12).
Reason. An RT image may only be used while it is temporally accurate. [L] p. 116–120 (temporal accuracy, phase sensitivity), 146–148 (permanence).
Acceptance. Replay delayed observations, a stream of targets based on old frames, expired targets, lost sync and a clock step: invalid inputs never drive the laser, a healthy pair of cameras can still track, and the local target timeout is not extended by a clock change. Boundary tests use the limits agreed in OQ-12.

F-VLT-23 Receivers check every message between camera, control and galvo nodes. A message with the wrong format or version, an unknown sender, or non-finite values can't update the accepted state or refresh a timeout. The same holds for duplicates, out-of-order messages and messages from before the sender's restart. Under a burst, memory stays bounded, and old messages are dropped, not processed late.
Reason. Independently developed nodes must agree on what is current; a stale message must never look fresh. [L] p. 137 (communication errors), 146 (permanence), 237 (idempotent messages).
Acceptance. Contract tests inject wrong versions or senders, truncated messages, NaN/∞, duplicates, reordering and a sender restart; only current valid inputs update the state. A burst beyond capacity: memory bounded, expired messages discarded per F-VLT-22.

F-VLT-24 Before arming, configuration and calibration are validated together. Units and coordinate convention, camera identities and image modes, finite parameters and cross-field limits must agree, and all nodes must use the same setup. Missing, corrupt or incompatible files keep the laser off and produce a specific diagnostic (local log and monitoring, F-VLT-40). Nodes don't combine data or targets from different setups.
Reason. Wrong calibration looks like a valid measurement but produces a wrong aim. [L] p. 162 (error detection), 289 (information representation, units).
Acceptance. Remove or corrupt a file, swap camera calibrations, change the image mode or the setup on one node, set invalid limits: arming is refused with a diagnostic; incompatible peer data is rejected. A matching validated setup passes F-VLT-3/5 after restart.

F-VLT-25 The control node knows the actual laser state and the reason for laser-off, as reported by the galvo node, within 50 ms of every change and at least every 50 ms. It uses the actual state, not its own command, for F-VLT-16 and hit counting. When the galvo node restarts, the control node enters the safe state, so a restarted galvo node never switches the laser on without re-arming.
Reason. The effect at the actuator is the evidence, not the target; a reintegrated galvo node must not resume an old armed state. [L] p. 206–207 (end-to-end protocol), 240 (reintegration).
Acceptance. Restart the galvo node while armed and tracking: laser stays off until re-armed; the display shows the galvo restart. Force each laser-off reason: reported within 50 ms. Stop the status reports: control node enters the safe state within 100 ms + 10 ms.

F-VLT-26 While armed, the control node monitors the effect of the actuator. If the spot is seen more than 20 mm outside the cube volume, the system switches the laser off within 100 ms without waiting (T3), raises an alarm and enters the safe state (latched, F-VLT-21). If the aiming error stays above 50 mm for 500 ms while the ball is tracked and the laser is on, counted from the exposure of the first frame above 50 mm, the system does the same within a further 100 ms (T6, ≤ 600 ms in total). A measured aiming error ≤ 50 mm ends the episode; a missing or unmeasurable error does not end it and is handled by F-VLT-42. Both values are configurable; they are set so that normal tracking (NF-VLT-1) never trips them.
Reason. A blocked or misaimed actuator must be detected by its effect. [L] p. 206–207 (end-to-end protocol), 247 (fault-tolerant actuators).
Acceptance. Block one mirror mechanically (or load a calibration with an offset in a test profile) while armed: laser off and alarm within 500 ms + 100 ms of the first frame above 50 mm. Spot projected outside the cube by a second pointer: alarm and laser off within 100 ms, without a 500 ms wait. An error ≤ 50 mm before 500 ms ends the episode; frames without a measurable error do not. Three normal runs: no false trip.

F-VLT-27 Before an observation is used for the ball state, the control node checks its plausibility. With three cameras, a camera whose observation disagrees with the other two by more than 10 px (reprojection, configurable) is excluded for that update and counted as not delivering. A ball state outside the cube volume is not aimed at. No pendulum motion model is used. Rejected updates count as gaps, never silently as hits.
Reason. A value failure of one camera must not move the aim; with three cameras it can be outvoted. [L] p. 230–232 (TMR, voters).
Acceptance. Replay with one camera reporting a false ball (synthetic offset of 30 px): that camera is excluded and the 3D error stays within F-VLT-3 limits; a ball state outside the cube is not aimed at; rejection counts are reported per run.

F-VLT-28 The nodes can be powered and started in any order. Each node starts in a defined ground state: laser off, mirrors at centre, unarmed, no ball state, no history from a previous session. The system reaches “ready to arm” without manual sequencing within 60 s after the last node is up, including clock synchronization and camera calibration, when a valid galvo calibration is stored. Without one, the galvo scan (F-VLT-13) follows, and “ready to arm” is reached when a scan succeeds: within 60 s plus 60 s per scan attempt (at most 2 attempts). If both attempts fail, the system ends in the safe state with the laser off and a diagnostic (F-VLT-13) within the same bound. During start-up the laser is on only while that scan runs (F-VLT-8). A controlled shutdown switches the laser off and centres the mirrors before the galvo node stops.
Reason. Start and restart must be defined; the ground state is the safe point for start-up and reintegration. [L] p. 79 (ground state), 186 (mode changes), 240.
Acceptance. Power up the five nodes in three different orders with a valid galvo calibration stored: ready to arm in every case within 60 s after the last node is up; laser off throughout (photodiode). Power up without a galvo calibration: the laser is on only while the scan runs (photodiode against log), and ready to arm within 120 s after the last node is up when the first scan succeeds. First scan forced to fail, second succeeds: ready to arm within 180 s. Both scans forced to fail: safe state, laser off and diagnostic within 180 s, no third scan. Documented shutdown: laser off before the mirrors stop being driven.

F-VLT-29 The control node and the galvo node log every mode and safety transition with the shared time and the node’s local time. Logged events: arm, disarm, safe state with reason, safety latch set or cleared, kill switch, degraded with camera, alarm, sync lost or regained, arming refused with reason, galvo scan started, failed or given up, setup invalid, and run start and end with result. Camera nodes log their own diagnostics (calibration failed, moved camera) the same way. For F-VLT-42 the log also holds the watchdog origin, the expiry and the time of the laser-off request. The log contains no images, is written without blocking the loop, and its size per session is bounded.
Reason. Every detected error must leave evidence for the fault-injection campaign and for faults without the laptop. [L] p. 254 (log column of the reaction matrix), 284 (fault injection).
Acceptance. After a fault-injection session every injected fault appears in the log with its reason, in the right order. Log writing throttled to the SD card’s worst case does not change loop timing (NF-VLT-18).

F-VLT-30 Software senses the kill switch (galvo node) and reports it; a broken sense wire reads as pressed. Pressing the kill switch puts the system into the safe state within 100 ms (ADR-VLT-007). The kill switch is the operator’s only physical control: pressing it disarms, and releasing it allows re-arming per F-VLT-35 (ADR-VLT-013). The supply interruption itself stays independent of software (NF-VLT-3).
Reason. The software must not stay armed across a kill-switch press, and the operator needs one deliberate physical control. [L] p. 217 (safety), 52–53 (hard deadline).
Acceptance. Press the kill switch during tracking: laser off immediately (hardware), display shows it within 100 ms, safe state; release: laser stays off until the control node re-arms after the 2 s hold-off. Unplug the sense wire: reported as pressed.

F-VLT-31 Before every arming the galvo node projects 3 check points onto floor or walls (points hidden by the ball are replaced) and the cameras measure them. If any lies more than 5 mm from its commanded position, a point can’t be measured, or F-VLT-42 expires, arming is refused with the laser off, a diagnostic names the reason, and the control node starts a galvo scan by itself (F-VLT-13) (ADR-VLT-012/013). The check takes ≤ 1 s with the laser on inside the cube and runs without an operator.
Reason. A moved galvo would aim wrongly at the ball; the error must be detected before the laser is used. [L] p. 162 (error detection), 206.
Acceptance. Shift the galvo mounting slightly (or load a calibration with an offset): arming refused with the diagnostic, a scan starts by itself, afterwards the check passes and the system arms. Hide a check point so that it can’t be measured: the check fails, the system does not arm. Correct calibration: arming succeeds and the check takes ≤ 1 s (photodiode, log).

F-VLT-32 On request in the configuration, the control node records a session: every observation, target, galvo status report, mode event and timestamp the loop uses, no images. Recording never blocks the loop and its size per session is bounded. A recorded session can be fed into the hardware-free replay (NF-VLT-22) and yields the same decisions.
Reason. Faults seen on the real setup must be reproducible offline, without the probe effect of extra instrumentation. [L] p. 281 (cluster simulation), 282 (probe effect).
Acceptance. Record a 10-min session and replay it offline: mode changes, validity decisions and run results identical; NF-VLT-18 unchanged with recording on; recording stops at its size limit without affecting the loop.

F-VLT-33 Two dry-run modes are set in the configuration. In *laser disabled*, the whole loop runs and the mirrors move, but the laser stays off. In *galvo disabled*, only tracking and monitoring run, with no galvo output. The display shows the active mode.

Reason. The loop must be testable during integration without laser risk. [L] p. 267–269 (design for testability, test doubles).
Acceptance. In each mode a full run produces tracking and monitoring data; a photodiode shows the laser never on; in *galvo disabled* the galvo output stays at centre.

F-VLT-34 The monitoring laptop shows live: mode (F-VLT-41), current aiming error, hit rate of the current and last run, camera status per node (F-VLT-11), galvo status and laser-off reason (F-VLT-25), clock offset per node and worst pairwise offset (F-VLT-18), and the camera images with the detected ball and spot marked.
Reason. The operator's only status display (F-VLT-40). Monitoring is a soft real-time function (T4). [L] p. 52 (soft deadline), 128 (composability).
Acceptance. During a run every listed value is visible and updates at least once per second; images at ≥ 1 per second per camera.

F-VLT-35 Arming and disarming are decided by the control node software (ADR-VLT-013). The only physical operator control is the kill switch (NF-VLT-3, F-VLT-30); there is no arming button. The control node arms when all of these hold: the kill switch is released, no safety fault is active or latched (F-VLT-21), the setup is valid (F-VLT-24), ≥ 2 cameras have seen the ball continuously for 2 s, and the arming check passes (F-VLT-31). The same rule applies at start and after a safety fault. The system disarms on every safety fault and when the kill switch is pressed. No network command or other control path arms, disarms or starts a calibration.
Reason. One defined, local control path; the laser comes on only with all prerequisites met. [L] p. 186 (mode changes), 217 (safety).
Acceptance. After start and after every safety fault: the laser switches on only after ≥ 2 cameras have seen the ball for 2 s and the arming check has passed; with the kill switch pressed or the setup invalid it never arms. Pressing the kill switch disarms. No network command arms or disarms (NF-VLT-10); the setup has no physical operator input other than the kill switch (inspection).

F-VLT-36 Dropped by the sponsor at the PM review (2026-11-02) — was: "recalibration recommended" hint on a systematic aiming error.
Reason. Replaced by the arming check, which starts a galvo scan by itself (F-VLT-31).
Acceptance. —

F-VLT-37 A system health overview on the monitoring laptop shows per node: status, software version, setup identity, clock state (synchronized, offset, clock stepped or stopped since start), CPU and RAM, message rates in and out, dropped and rejected messages, and the latency per stage of the tracking chain (median and 99th percentile over the last second; full statistics in F-VLT-39). Every value that has a limit in these requirements is shown with that limit and highlighted when outside. Highlighted too: different versions or setups between nodes, and a clock that stops, jumps or runs at a different rate than the others.
Reason. Integration faults must point to the node and stage that cause them. [L] p. 163 (monitoring the message flow), 266 (monitor the budget in operation).
Acceptance. Inject one fault each — unsynced camera, clock step on one node, frozen node process, overloaded camera node, wrong software version, delayed messages: within 2 s the highlighted value names the faulty node and stage. All values update at least once per second; NF-VLT-6/8 still hold.

F-VLT-38 The fused 3D ball position, the predicted aim point and the measured spot are drawn into every camera image on the monitoring laptop, next to that camera's own detection.
Reason. A wrong calibration becomes visible at a glance. [L] p. 162 (error detection).
Acceptance. Shift one camera's calibration by 5 px: its image shows the drawn position visibly off the detected ball while the other two match; with a correct setup drawn and detected positions agree within the F-VLT-3 tolerance.

F-VLT-39 Chain delay statistics. For every position update the delay of each stage of the tracking chain is traced: capture and readout, detection, camera → control transmission, waiting for the position update, fusion and prediction, control → galvo transmission, waiting until placement, galvo output. Stages that run per camera are traced per camera. Mirror settling, which the nodes can't see, comes from the independent measurement (NF-VLT-29). For each stage and end to end, the monitoring laptop shows count, minimum, maximum, mean, standard deviation (jitter), median, 95th and 99th percentile. Three windows: last second, current run, whole session (resettable on the laptop). It also shows each stage's share of the end-to-end delay and a histogram with the 30 ms limit (NF-VLT-18) and each stage budget marked. Deadline misses and discarded targets are counted. At the end of each run and session the statistics are saved on the laptop as a file (numbers only) for the test report.
Reason. The deadline is met only if every stage stays in its budget; maxima and their distribution must be visible. [L] p. 39–40 (budget per stage), 68 (action duration statistics), 266 (testing the temporal domain).

Acceptance. 10-min run: statistics for all stages and windows shown and saved; the sum of stage means equals the end-to-end mean within 1 ms; adding a 5 ms delay in one stage (test profile) shows in that stage's statistics and share within 2 s, not in the others; values agree with the logged per-stage timestamps of NF-VLT-18.

F-VLT-40 Operating the demo requires the monitoring laptop: it is the operator's only status display, and no node has a local indicator (sponsor decision 2026-11-10). Without the laptop the system still tracks and stays safe (NF-VLT-7), but the operator sees only the laser. Therefore: (a) within 2 s of connecting or reconnecting, the laptop shows the current state of every node. That covers mode, armed or safe, an active safety latch with its cause, the kill switch, and the reason the system is not armed. It also shows the diagnostics and alarms since the node started, not only events after the connection. (b) A node from which no monitoring data has arrived for 2 s is shown as stale, and its last values are marked as not current. (c) Every diagnostic and alarm shown on the laptop is also written to the local log (F-VLT-29).
Reason. The only display must never show an old state as current. [L] p. 52 (soft deadline), 163 (monitoring the message flow).
Acceptance. Start without the laptop, inject an alarm and an invalid setup on one camera, then connect the laptop: within 2 s it shows the latch with its cause, the setup error naming the camera, and why the system is not armed. Stop one node's monitoring sender: shown as stale within 2 s. After a session without the laptop every injected fault is in the log.

F-VLT-41 The system has these operating modes, shown on the display (F-VLT-34) and logged (F-VLT-29). **Start** (ground state, calibrating cameras, synchronizing clocks). **Calibrating galvo** (F-VLT-13). **Ready** (not armed, all prerequisites met except the ball). **Arming check** (F-VLT-31). **Tracking**, normal or degraded (F-VLT-11). **Safe**, latched or not (F-VLT-21). The configurations are: normal, dry run *laser disabled*, dry run *galvo disabled* (F-VLT-33), with or without session recording (F-VLT-32). Changes towards laser off take effect within the deadlines of NF-VLT-26. The laser can come on only through *arming check* → *tracking*, or in *calibrating galvo*. A configuration takes effect only at a node start. No other mode or transition exists.
Reason. Modes and mode changes must be listed so that every transition is tested; switching configuration at runtime is unsafe. [L] p. 186 (mode changes take effect at the ground state), 287 (modes of operation and configurations).
Acceptance. In the fault-injection campaign (NF-VLT-23) and three normal runs, the logged mode sequence contains only the listed transitions; the laser is never on in a mode other than *arming check*, *tracking* or *calibrating galvo* (photodiode against log); changing the configuration of a running node has no effect until restart.

F-VLT-42 Whenever the laser is on (galvo scan, arming check, tracking), the control node checks that the spot is measurable. From the first laser-on placement, a measurable spot must be seen within 500 ms and then at least every 500 ms while the laser stays on. While tracking, "measurable" means a valid aiming error (F-VLT-6); during a scan or check it means a spot located in 3D by at least two cameras, without a ball. A missing spot, a spot off the ball seen by one camera only, and stale or rejected observations do not count. The interval is measured on a local monotonic clock; clock corrections, new targets and mode changes with the laser on don't restart it. When it expires the laser is off within 100 ms (T6, ≤ 600 ms in total), and a later measurement can't cancel this. Tracking: alarm and safe state (latched, F-VLT-21). Galvo scan: the scan fails (F-VLT-13). Arming check: the check fails (F-VLT-31). A faster shutdown (ball lost, sync loss, kill switch, output stop) always wins. The value 500 ms is configurable for test profiles only (ADR-VLT-018/019).
Reason. The laser must not stay on when its effect can't be measured: F-VLT-26 only reacts to a measured error, so a spot that can't be measured had no time limit. [L] p. 206–207 (end-to-end protocol, effect monitoring), 254 (reaction matrix).
Acceptance. Replay and fault tests: no spot from the first emission, spot lost later, spot off the ball seen by one camera, stale or duplicate observations, a clock step and a target change during the interval, a valid measurement just before and just after expiry. In each case the laser is off ≤ 600 ms after the stated origin (photodiode, NF-VLT-29), with the outcome of the current mode; invalid data never extends the interval. No alarm while the laser is actually off.

3. Non-functional requirements

NF-VLT-1 Acceptance of the demo: in a run (F-VLT-7), the hit rate from release until the amplitude is below 50 mm is at least 80 %. This holds with all three cameras and with each of the three two-camera pairs (degraded mode; sponsor, OQ-14). **Operating envelope** (OQ-14, decided 2026-11-12): the ball (≈ 5 cm on a ≈ 1 m string) is released by hand from a horizontal deflection of 100 … 300 mm from its rest position; ball speed up to 1000 mm/s and acceleration up to 4000 mm/s²; ordinary room lighting: the room's normal ceiling lights, no extra lamps and no direct sunlight on the cube.
Reason. The demo's purpose, stated as a measurable goal. [L] p. 20 (falsifiable acceptance).

Acceptance. System-reported hit rate ≥ 80 % in 3 of 3 consecutive runs with three cameras and in 3 of 3 consecutive runs for each camera pair (third camera stopped); each set of three runs is released from ≈ 120 mm, ≈ 200 mm and ≈ 300 mm (measured by the system at release; F-VLT-7 starts a run only above 100 mm); an aborted run counts as failed; the lighting (lights on, no extra lamps, no direct sunlight on the cube) is recorded with each run.

NF-VLT-2 Laser safety (IEC 60825-1): the laser is visible and at most Class 2 (≤ 1 mW). The student team is responsible for laser safety in the lab (OQ-6). Before the first laser-on test it records a laser-safety procedure that also covers laser-on without an operator action (arming check, galvo scan, ADR-VLT-013).
Reason. Safety requirement of a lab demo with a laser. [L] p. 217 (safety).
Acceptance. Given laser: Laserfuchs LFD650-1-12, 650 nm, 1 mW, Class 2 per manufacturer (ADR-VLT-015). The label on the delivered module shows Class 2 or lower; recorded in the setup documentation together with the team's laser-safety procedure and the responsible team role.

NF-VLT-3 A physical kill switch cuts the laser supply independent of all software.
Reason. When every electronic layer fails, a path that needs no computer remains. [L] p. 248.
Acceptance. Kill switch pressed during tracking: laser off immediately, with all nodes still running.

NF-VLT-4 Dropped by the sponsor at the PM review (2026-11-02) — was: external hardware watchdog; replaced by NF-VLT-30.
Reason. See NF-VLT-30.
Acceptance. —

NF-VLT-5 All nodes share one clock synchronized over Ethernet (software clock synchronization, ADR-VLT-002) with an offset ≤ 1 ms between any two nodes; every observation carries its exposure time in that clock.
Reason. Observations from different cameras must refer to comparable physical times. [L] p. 82 (why clock synchronization), 92 (precision π).
Acceptance. The worst pairwise offset per F-VLT-18 stays ≤ 1 ms during a 10-min run (each node ≤ 0.5 ms to the clock reference). Cross-check: one LED flash seen by all cameras yields exposure times within 1 ms + one frame period.

NF-VLT-6 Sending monitoring data never blocks the tracking loop; under back-pressure monitoring data is dropped, not queued.
Reason. Monitoring must not change the timing of the loop. [L] p. 128 (composability), 161 (non-blocking write).
Acceptance. Monitoring server paused (SIGSTOP) during a run: loop timing (NF-VLT-18) unchanged, node memory does not grow.

NF-VLT-7 Tracking and safety behave identically with and without the monitoring laptop connected.
Reason. Safety must not depend on an observer. [L] p. 128 (composability).
Acceptance. NF-VLT-1 and F-VLT-8 … F-VLT-11 pass with the laptop absent.

NF-VLT-8 Monitoring traffic of all nodes together stays ≤ 20 Mbit/s; camera images are sent at ≤ 5 per second, ≤ 320 × 240 px.
Reason. Monitoring shares the network with the loop and clock synchronization. [L] p. 60 (load hypothesis), 138 (thrashing).
Acceptance. Switch port counters or capture over a 10-min run: total monitoring traffic ≤ 20 Mbit/s.

NF-VLT-9 Loop timing (NF-VLT-18) and the demo acceptance (NF-VLT-1) are verified both with monitoring on (at its maximum rate) and off.
Reason. Instrumentation can change what it measures. [L] p. 128, 282 (probe effect).
Acceptance. Both measurements recorded in the test report.

NF-VLT-10 No network message controls the running loop. The loop processes on every node have no command interface, and nothing over the network arms, disarms, starts a calibration or changes running behaviour. The monitoring software is observe-only (ADR-VLT-003). Separately, admin access is allowed (ADR-VLT-014): authenticated SSH, key only with no password login, from devices on the demo network (NF-VLT-28). It covers installing and updating software, starting, stopping and restarting node software, reboot and shutdown, editing configuration and calibration files, and fetching logs, recordings and statistics. Configuration and calibration changes take effect only when the node software restarts, and that restart follows the existing rules (F-VLT-8/17/21).
Reason. Security: only the defined local path controls the laser; the team still needs to administer five headless nodes. [L] p. 218 (security).
Acceptance. Port scan of every node: listening ports only for SSH, clock synchronization and the loop's own messages, which accept only loop messages from their expected senders (F-VLT-23). SSH with a password is

refused. Editing the configuration of a running node: behaviour unchanged until restart. Restarting the galvo software over SSH while armed: safe state, re-arming per F-VLT-35.

NF-VLT-11 Camera images are shown live only and are not stored by default (GDPR).
Reason. Personal data may appear in camera images.
Acceptance. After a run, no image files on any node or on the laptop unless storage was explicitly enabled in the configuration. Exception: the reference data set (NF-VLT-21).

NF-VLT-12 The system runs continuously for 10 minutes (several runs) without a restart.
Reason. A presentation consists of several runs in one session. [L] p. 266 (high-water mark over long runs).
Acceptance. 10-min soak: no node restart, no memory growth > 10 %, NF-VLT-1 still met in the last run.

NF-VLT-13 Every node can be set up reproducibly from the repository by a script. Code is version-controlled and reviewed before merge, and every node's code has automated tests.
Reason. Project standard: versioning, frozen tools. [L] p. 289.
Acceptance. A fresh SD card set up only with the documented script runs the node; the test suite runs with one target.

NF-VLT-14 The source code has one module per node type (camera, control, galvo) and one for the monitoring server.
Reason. Project standard: well-structured project documents and sources. [L] p. 289.
Acceptance. Inspection of the source tree.

NF-VLT-15 All nodes are Raspberry Pi 3 on one wired Ethernet switch (given hardware).
Reason. Given structure of the components. [L] p. 287 (given items are requirements).
Acceptance. Setup documentation lists the hardware; all tests run on it.

NF-VLT-16 The galvo node switches the laser on and off in software by switching the laser's supply (the given module has no TTL input, ADR-VLT-015). The laser housing is at supply potential, so the laser is mounted electrically insulated.
Reason. The software laser-off rules need an actuator; a live housing must not short or energize the frame. [L] p. 247, 217 (safety).
Acceptance. Galvo node toggles the laser 10 × per second for 10 s; photodiode confirms every transition. Insulation check: no continuity between the laser housing and the mounting frame.

NF-VLT-17 Laser colour, ball colour and background are chosen so that every camera can tell the spot apart on the ball and the ball against the background.
Reason. Observation quality bounds all later accuracy. [L] p. 111–112 (observations).
Acceptance. Proven by F-VLT-1 and F-VLT-2 passing on the reference data set, evaluated per camera. (laser fixed: red, ADR-VLT-015; open: OQ-7)

NF-VLT-18 End-to-end tracking transaction T1 (firm, ADR-VLT-008): from the exposure of a frame to the spot being placed at the corresponding aim, p99 ≤ 30 ms. The stages are capture/readout, detection, transmission, fusion and prediction, transmission, galvo output and mirror settling. Their individual budgets are chosen in the architecture ([L] p. 287: chosen items belong there) and must sum to ≤ 30 ms. A target whose validity has expired on arrival is discarded and counts as a miss.
Reason. Every distributed transaction needs an end-to-end deadline with a budget per stage. [L] p. 36–40 (transaction, budget), 52–53 (firm deadline).
Acceptance. Per-stage timestamps logged over a 10-min run plus the independent measurement of NF-VLT-29: 99th percentile of the end-to-end time ≤ 30 ms; maximum and count of discarded targets reported; each stage within its budget.

NF-VLT-19 Each node stays within ≤ 75 % average CPU and ≤ 50 % RAM on the Pi 3 during a run.
Reason. Resource requirements with headroom for peak load. [L] p. 55 (resource adequacy), 288 (resources).
Acceptance. Resource log during a 10-min run.

NF-VLT-20 Clock precision (NF-VLT-5) holds with monitoring traffic at its maximum (NF-VLT-8).
Reason. Network load adds latency jitter to clock synchronization. [L] p. 105 (latency jitter).
Acceptance. Clock offsets ≤ 1 ms during a 10-min run with monitoring at maximum rate.

NF-VLT-21 A reference data set is recorded once with the final setup. It contains time-synchronized frames of all three cameras covering: a resting ball at ≥ 10 positions spread over the cube, swinging runs (release until amplitude < 50 mm), the spot on and off the ball, laser off, frames without the ball, and the ball being carried outward by hand

(F-VLT-7). Every frame is annotated with the image positions of ball (centre and radius) and spot, in the same pixel coordinates as the detection (origin top-left, v downward). For the resting positions the 3D ball position is measured by hand. The set and its annotations are stored in the repository and are the input of the automated detection, triangulation, aiming-error, prediction and run tests. It is an explicit exception to NF-VLT-11, on condition that no people appear in the frames. After a significant setup change (camera, background, ball, laser or lighting) it is recorded again.
Reason. Detection and evaluation must be testable repeatably without hardware. [L] p. 264–266 (unit tests), 283 (test selection).
Acceptance. The data set exists in the repository with an annotation file that passes a schema check, covers every listed case for all three cameras, and contains no people (reviewed).

NF-VLT-22 The shared processing and decision logic can be tested without cameras, Pis or galvo hardware. Tests replay the reference data set and synthetic message and time events through the same logic used on the nodes, with a simulated clock and captured actuator output. Replay includes two-camera combinations, loss, delay and reordering, node restart and sync loss. It adds no live target path to the monitoring laptop.
Reason. Integration faults must be found before the real setup is available. [L] p. 269 (test doubles), 281 (cluster simulation).
Acceptance. One documented target runs the replay suite without hardware. The same input and event schedule always gives the same validity decisions, state transitions and run result; numeric outputs meet the existing tolerances. Fault cases check F-VLT-22 … 24. Physical timing and aiming acceptance still require the real setup.

NF-VLT-23 **Fault hypothesis.** Fault-containment regions: each of the five nodes, the Ethernet switch, converter board + galvo driver, the laser module. **Tolerated (k = 1, 2-of-3):** crash, omission or timing failure of one camera node, including its restart (F-VLT-11/17), and a detectable value failure of one camera while three deliver (F-VLT-27). **Detected and turned into the safe state (fail-safe):** crash or omission of the control node or the switch (F-VLT-10), loss of the clock reference or sync loss (F-VLT-20/22), galvo node restart (F-VLT-25), interrupted galvo output stream (F-VLT-15), converter link fault that produces an observable effect (spot outside the cube: T3; persistent aiming error or no measurable spot: T6; F-VLT-26/42, ADR-VLT-015/018), a second camera fault, actuator misbehaviour (F-VLT-16/26/42), galvo node software hang (NF-VLT-30, T7, within 1 s). **Not covered — residual risk, kill switch only (NF-VLT-3):** a hang shorter than 1 s (laser frozen at its last clamped aim), undetected consistent value failures of two cameras, a laser stuck on in hardware, a converter link fault without an observable effect. **Fault frequency:** at most one fault at a time; a second fault no earlier than 10 s after recovery from the first.
Reason. Guarantees hold only for faults the hypothesis covers. [L] p. 60 (fault hypothesis), 229 (node as unit of failure), 234 (fail-silent), 284 (fault injection beyond the hypothesis).
Acceptance. Fault-injection campaign on the real setup per NF-VLT-34, plus at least two faults beyond the hypothesis (two cameras at once; camera fault during control restart): each tolerated fault keeps tracking, each detected fault ends in laser off within its deadline (NF-VLT-26), each beyond-hypothesis case ends in laser off or is recorded as residual. Converter faults are injected both with and without an observable effect; the latter are recorded as outside the detection guarantee. Results logged per F-VLT-29.

NF-VLT-24 **Global time base.** The precision $\pi \le 1$ ms of NF-VLT-5 means that every node within $\pi/2 = 0.5$ ms of the clock reference. The global granularity g for comparisons between nodes is fixed with $g > \pi$ (reasonableness condition). Timestamps keep ns resolution, but the order or duration of events on different nodes is only claimed beyond $\pm 2g$. The fusion error budget (OQ-14) includes the timestamp uncertainty × maximum ball speed. Corrections of the shared clock never lengthen or shorten local timeouts (F-VLT-10, F-VLT-22). No other time service adjusts the node clocks.
Reason. Timestamps from different nodes are only comparable within the limits of time measurement. [L] p. 92 (precision), 97 (reasonableness $g > \pi$), 98 ($\pm 2g$), 106 (clock correction).
Acceptance. Configuration check on every node: only the shared clock synchronization adjusts the clock. Stop the clock reference during tracking: nodes flag unsynced within the sync-status age (OQ-12) and the laser goes off. Step the shared clock by 50 ms on the galvo node: the target timeout still fires at 100 ms ± 10 ms. The value of g is recorded once OQ-1/OQ-12 fix the camera timestamp source.

NF-VLT-25 **Fail-safe laser enable.** The laser emits only while the galvo node actively drives its enable signal. An unpowered, booting, rebooting or crashed galvo node, a broken enable wire or a floating input all leave the laser off, by hardware design.
Reason. The ground state of the actuator must be safe without software. [L] p. 79 (ground state), 217 (safety).
Acceptance. With the laser supply on and the kill switch closed: power the galvo node off, reboot it, kill the galvo process, unplug the enable wire — a photodiode shows the laser off in every case, including during the whole boot.

NF-VLT-26 **Real-time transactions and deadline classes** (sponsor, ADR-VLT-008). **T1 tracking** (exposure ⟶ spot placed) is *firm*, with the deadline and the handling of expired targets of NF-VLT-18. **T2 target-loss shutdown** (last valid target ⟶ laser off) is *hard* at 100 ms + 10 ms (F-VLT-10). **T3 unsafe-condition shutdown** (< 2 cameras, sync loss, alarm, actuator fault ⟶ laser off) is *hard* at 100 ms from the first frame exposure that shows the condition, or from the local event for faults detected on a node; it contains no 500 ms qualification. **T6 effect supervision** (aiming error > 50 mm for 500 ms, F-VLT-26; no measurable spot for 500 ms, F-VLT-42 ⟶ laser off) is *hard* at 500 ms + 100 ms from the origin stated there. **T7 galvo software hang** (output loop stops ⟶ laser off) is *hard* at 1 s in total, including reset and laser disable (NF-VLT-30). A faster applicable shutdown always wins. **T4 monitoring** is *soft*, best effort (NF-VLT-6). **T5 camera reintegration** has no deadline. The OS is not real-time, so hard deadlines are met by the software design, backed by the kill switch, and verified by measuring the maximum, not proven.
Reason. Deadlines are classified by the consequence of a miss. [L] p. 52–53 (soft, firm, hard), 266 (maximum by measurement).
Acceptance. 10-min run with monitoring at maximum (NF-VLT-9) plus the fault campaign (NF-VLT-34): T1 p99 and maximum reported, late targets counted; T2, T3, T6 and T7 maximum ≤ their deadline in every injected case, measured from their stated origin and independently (NF-VLT-29).

NF-VLT-27 **State semantics and load hypothesis.** Observations, targets and galvo status reports describe a state: only the newest valid one is used, older ones are skipped and never queued, and latency doesn't grow with load. Load hypothesis per channel at peak: camera ⟶ control: frame rate × observation size per camera; control ⟶ galvo: update rate × target size; galvo ⟶ control: status rate; clock synchronization: sync rate; monitoring ≤ 20 Mbit/s (NF-VLT-8). In total ≤ 50 % of each node's 100 Mbit/s link. The rates are fixed when OQ-1/OQ-12 fix the frame and update periods.
Reason. Response times can only be guaranteed below a peak load; a system must saturate, not thrash. [L] p. 60 (load hypothesis), 130 (state messages), 138 (thrashing).
Acceptance. Load test at 1.5 × the nominal camera frame rate (synthetic observations) with monitoring at maximum: end-to-end latency of the newest frame stays bounded (no growing backlog), skipped frames are counted, node memory constant. Port counters confirm each channel within its hypothesis.

NF-VLT-28 **Isolated network and minimal node platform.** The five nodes and the monitoring laptop are the only devices on a dedicated switch, with no uplink to the lab network or the internet. For administration (NF-VLT-10) a team machine may also be connected, but not during acceptance runs. The network is trusted, so senders are identified without cryptographic authentication. Each node runs a minimal OS image set up by the script of NF-VLT-13: no desktop, Wi-Fi and Bluetooth off, no automatic updates, no time service other than the shared clock synchronization.
Reason. Hidden channels through shared resources and outside traffic must be excluded; security by isolation. [L] p. 128 (composability), 218 (security).
Acceptance. During acceptance runs the switch port list shows only the six devices; no default route on any node. Setup-script check on a fresh SD card: listed services disabled. NF-VLT-18 measured with this image.

NF-VLT-29 **Independent timing measurement.** End-to-end timing (T1) and the shutdown deadlines (T2, T3, T6, T7) are verified with an instrument independent of the node clocks, not only from the nodes' own timestamps. For example: an LED flash in the cameras' view timed by a separate microcontroller or scope, and a photodiode at the galvo output or on the ball.
Reason. Self-reported timestamps share the errors of the system they measure. [L] p. 266 (testing the temporal domain), 282 (probe effect).
Acceptance. The test report contains, for T1, T2 and T3, the independent measurement (≥ 100 samples each) next to the self-reported values; their difference is stated. For T6 and T7, ≥ 10 injections per case, with fault injection, origin and laser off recorded separately.

NF-VLT-30 If the galvo node's output loop stops (software hang), the laser goes off within 1 s in total (T7, including reset and laser disable), by the Pi's on-chip watchdog fed by the output loop (sponsor, ADR-VLT-010). The reset switches the laser off by NF-VLT-25, and the node reboots into the ground state (F-VLT-28).
Reason. A hung output task can keep the laser on; the heartbeat belongs at the end of the output task. [L] p. 167 (watchdog).
Acceptance. Freeze the galvo output loop (SIGSTOP) while the laser is on: photodiode shows the laser off within 1 s of the hang, including the reset; the node reboots into the ground state.

NF-VLT-31 Each node type can run on the real setup while any of its peers is replaced by a simulator sending the same messages. The simulators are: a camera simulator playing recorded data, a galvo simulator recording the targets it receives, and a control simulator sending scripted targets. Simulators never drive the laser.

Reason. Nodes are integrated one at a time against test doubles. [L] p. 269 (test doubles), 279–281 (integration testing).
Acceptance. Each node passes its node tests with all peers simulated and with any mix of real and simulated peers; the health overview (F-VLT-37) marks simulated nodes.

NF-VLT-32 **Physical layout for measurable hits**. The galvo is placed so that, throughout the operating envelope (NF-VLT-1), every pair of cameras includes at least one camera that sees the spot on the ball. The floor and walls used by the galvo scan and the arming check (F-VLT-13/31) are each visible to at least two cameras and don't block any camera's view of the tracking volume. Seen from the galvo, the whole cube lies within the usable optical scan angle with ≥ 10 % margin. The usable angle depends on the DAC level, the driver gain and any amplifier (ADR-VLT-015, OQ-4).
Reason. NF-VLT-1 must hold for every camera pair, and a hit can only be counted when a camera sees the lit side of the ball. Given hardware geometry is a requirement. [L] p. 287.
Acceptance. On the reference data set (NF-VLT-21): for every camera pair, at least one camera of the pair sees the spot in ≥ 95 % of the frames of the swinging runs with the spot on the ball. Every scan point of F-VLT-13 is seen by ≥ 2 cameras. Commanding the cube's corners: every corner is reached with the DAC output ≤ 90 % of its full scale.

NF-VLT-33 **Timing of actions.** Every periodic action has a specified minimum and maximum period and a worst-case execution time: frame capture and detection per camera, position update and prediction, galvo output, galvo status report, monitoring. The worst-case execution time is measured on the target under the load hypothesis (NF-VLT-27) as the high-water mark over a 10-min run. Each action meets execution time < period. Every RT image that is used meets the phase-sensitivity condition update delay + period + execution time < temporal accuracy (F-VLT-22). We choose the period values, so they are set in the architecture.
Reason. Deadlines can only be met if every action has a bounded period and execution time. [L] p. 65 (d_WCET, d_MINT), 72 (WCET), 120 (phase-sensitive RT images), 266 (high-water mark), 287 (timing of actions).
Acceptance. For every listed action: measured period minimum and maximum within the specified bounds and execution-time high-water mark below the period over a 10-min run with monitoring at maximum; the phase-sensitivity condition holds with the measured values.

NF-VLT-34 **Error detection coverage and reaction matrix.** Every error in the fault hypothesis (NF-VLT-23) is covered by the reaction matrix below: what detects it, how the system reacts, within which time, and what is logged (F-VLT-29). **Coverage target:** 100 % of the injected errors within the hypothesis are detected and handled within their time limit, with each matrix row injected at least 10 times on the real setup. Errors beyond the hypothesis, and faults without a guaranteed detection (NF-VLT-23), are injected and their outcome recorded separately; they don't count towards the coverage target.
Reason. Error detection needs numbers and mechanisms; every detectable error maps to a defined reaction. [L] p. 254 (reaction matrix), 284 (fault injection), 288 (error detection coverage).
Acceptance. Fault-injection report: per row the number of injections, detections and reactions within the time limit (target 10 of 10 for every row); beyond-hypothesis cases listed with their outcome.

3.1. Reaction matrix (NF-VLT-34)

Error	Detected by	Reaction	Time limit	Requirement
one camera crashed, silent or late	no valid observation within the age limit	camera absent, degraded; < 2 cameras → safe	observation-age limit (OQ-12); < 2 cameras: 100 ms (T3)	F-VLT-11, F-VLT-22
one camera reports a wrong ball (3 cameras)	plausibility against the other two	camera excluded for that update	next update	F-VLT-27
camera moved	marker check against the calibration	camera absent, diagnostic	1 s	F-VLT-12
ball seen by < 2 cameras	camera status	laser off; > 1 s ⟶ safe	100 ms / 1 s	F-VLT-11
control node, switch or network down	target timeout at the galvo node	laser off, safe	100 ms + 10 ms (T2)	F-VLT-10
galvo node restarted	restart reported by the galvo node	safe, re-arming needed	100 ms (T3)	F-VLT-25
galvo node software hang	on-chip watchdog	reset, laser off	1 s in total (T7)	NF-VLT-30
clock unsynchronized or sync status too old	offset and sync-status age	data of that node excluded; galvo or < 2 cameras ⟶ safe	100 ms (T3)	F-VLT-20, F-VLT-22

Error	Detected by	Reaction	Time limit	Requirement
malformed, duplicate, out-of-order or pre-restart message	receiver checks	message dropped	immediately	F-VLT-23
expired target or stale source data	validity check	target discarded, counted as miss	immediately	F-VLT-22, NF-VLT-26
galvo output stream interrupted	galvo node output monitoring	laser off, safe	100 ms (T3)	F-VLT-15
converter link broken (I2S clock or data) with an observable effect	the effect: spot outside the cube, aiming error or no measurable spot (clock loss also centres the mirrors by hardware)	as for that effect (rows below)	as for that effect (T3 or T6)	F-VLT-26/42, ADR-VLT-015/018
converter link broken without an observable effect	none guaranteed	residual; kill switch	no guaranteed time	NF-VLT-23
spot seen while the laser is off	camera observation vs. actual laser state	alarm, safe (latched)	100 ms (T3)	F-VLT-16
spot more than 20 mm outside the cube	spot position	alarm, safe (latched)	100 ms (T3)	F-VLT-26
mirror blocked or misaimed	aiming error > 50 mm	alarm, safe (latched)	500 ms + 100 ms (T6)	F-VLT-26
no measurable spot while tracking	optical-effect watchdog	alarm, safe (latched)	500 ms + 100 ms (T6)	F-VLT-42
no measurable spot during galvo scan or arming check	optical-effect watchdog	laser off, scan or check fails	500 ms + 100 ms (T6)	F-VLT-13/31/42
kill switch pressed or sense wire broken	sense contact	safe (laser cut by hardware)	100 ms	F-VLT-30
invalid or mismatched setup	validation before arming	arming refused, diagnostic	before arming	F-VLT-24
galvo calibration off	arming check	arming refused, galvo scan	1 s	F-VLT-31
repeated safety faults	fault counter	safe state latched	3 faults in 60 s	F-VLT-21
monitoring laptop absent or stalled	— (not needed)	none; loop unchanged	—	NF-VLT-6, NF-VLT-7

4. DRS requirements checklist

The lecture's checklist "Requirements for DRS" ([L] p. 287–288). Rule ([L] p. 287): an item that is given to us is a requirement; an item we choose belongs in the architecture and is deferred there with that reason.

#	Checklist item	Covered by	Deferred / still open (reason)
1	desired end-to-end functions at system level	F-VLT-1 … 7, NF-VLT-1	—
2	modes of operation and possible configurations	F-VLT-41, F-VLT-33, F-VLT-35	—
3	structure of all components — nodes, networks, gateways, I/O	NF-VLT-15, ADR-VLT-001, NF-VLT-28; no gateway (one cluster, the laptop is a sink)	—
4	interfaces — structure, timing, control mechanisms	given interfaces: camera, galvo driver, converter, laser, kill switch (given components, F-VLT-15/19/30, NF-VLT-16)	values open: OQ-4 (hardware facts; camera given in given components, laser and DAC fixed by ADR-VLT-015). Interfaces between our own nodes are chosen by us → architecture
5	messages and data elements — semantics, coding, size	semantics: F-VLT-1/2, F-VLT-23, NF-VLT-27	coding and size are chosen by us → architecture

#	Checklist item	Covered by	Deferred / still open (reason)
6	timing of actions — WCET, minimum and maximum period	NF-VLT-33, F-VLT-3 (fixed period), F-VLT-25 (50 ms)	period values are chosen by us → architecture, checked against NF-VLT-33
7	end-to-end deadline of each transaction, and its budget per stage	NF-VLT-18, NF-VLT-26 (T1 … T7), F-VLT-39	stage budgets are chosen by us → architecture; their sum ≤ 30 ms is required (NF-VLT-18)
8	temporal accuracy — how old each value may be when it is used	F-VLT-22, F-VLT-5, NF-VLT-33 (phase sensitivity)	values: OQ-12 (derived from tolerated error ÷ maximum ball speed, NF-VLT-1)
9	time base — the required precision of the global time	NF-VLT-5, NF-VLT-24, F-VLT-18, F-VLT-20	g: OQ-12 (camera timestamp source)
10	startup, restart and mode changes	F-VLT-8, F-VLT-17, F-VLT-25, F-VLT-28, F-VLT-41	—
11	resource requirements — memory, CPU, I/O	NF-VLT-19, NF-VLT-8, NF-VLT-27, F-VLT-29/32 (bounded storage)	—
12	fault-tolerance requirements — the fault hypothesis	NF-VLT-23, F-VLT-11, F-VLT-27	—
13	performance requirements — the load hypothesis	NF-VLT-27, NF-VLT-8, NF-VLT-1 (operating envelope of the process)	rates: OQ-12
14	failure modes of the nodes	NF-VLT-23 (crash, omission, timing, value per region)	—
15	error detection coverage — numbers and mechanisms	NF-VLT-34 (reaction matrix, 100 %, ≥ 10 injections per row), F-VLT-42	—
16	safety requirements	NF-VLT-2, NF-VLT-3, NF-VLT-25, NF-VLT-30, F-VLT-8 … 10, F-VLT-16, F-VLT-21, F-VLT-26, F-VLT-30	—
17	security requirements	NF-VLT-10, NF-VLT-28, NF-VLT-11	—
18	traceability — every item points to its check	every requirement has a falsifiable acceptance criterion and cites the decisions (ADR-VLT-n) and questions (OQ-n) it depends on	test cases with their own IDs, each pointing to the requirement it checks, are written in the test concept of M3 ([L] p. 20, phase 7) — the test phase follows the architecture

4.1. Given components and interfaces

Component	Direction	Given contract	Owner (role)
Raspberry Pi Camera Module v2.1 on each camera node (given hardware)	in	Sony IMX219, 8 MP (3280 × 2464), **rolling shutter**, fixed-focus lens 3.04 mm f/2.0, field of view 62.2° × 48.8° on the full sensor; 15-pin CSI-2 to a Raspberry Pi 3 Model B+; Linux camera stack (libcamera) on Raspberry Pi OS. Sensor modes (from the published mode list, to be confirmed on the delivered camera): 640 × 480 up to ≈ 103 fps on a cropped 1280 × 960 sensor area; 1640 × 1232 up to ≈ 42 fps and 3280 × 2464 up to ≈ 21 fps on the full area; 1920 × 1080 up to ≈ 48 fps cropped. Exposure timestamp semantics and measured timing must support NF-VLT-18 (OQ-12).	Sensor and actuator modules
Kernel PTP (software timestamping)	both	IEEE 1588 over UDP on the demo switch (ADR-VLT-002)	student team
Galvo scanner set with driver (AliExpress item 1005013402042428, “15Kpps Laser Galvo DIY SLA 3D”, ADR-VLT-015)	out	analog X/Y input ±5 V, 200 kΩ, 0.33 V/°; optical scan angle ±15° standard, up to ±30°; small-step response ≤ 0.5 ms; linearity 99.9 %; mirror 11 × 7 mm, > 99 % at 400–700 nm; supply ±12 V DC, 0.5 A average, 1.2 A peak; output chain tested on the bench 2026-11-12: galvo node → PCM5102A → driver → mirrors steers the laser, no pattern board in use, power supply from the set (tested);	Sensor and actuator modules

Component	Direction	Given contract	Owner (role)
		input wiring type, usable angle and measured response open (OQ-4)	
Converter board galvo node → galvo driver	out	PCM5102A I2S stereo DAC module (ADR-VLT-015): I2S from the galvo node; L = X, R = Y; DC-coupled, ground-centred, ≈ ±2.97 V single-ended; mutes to 0 V on clock loss; one-way (no read-back)	Sensor and actuator modules
Laser module	out	Laserfuchs LFD650-1-12 (ADR-VLT-015): 650 nm, 1 mW, Class 2, 3–12 V DC, 15–30 mA; two power wires, switched via its supply; housing at +V	Sensor and actuator modules
Kill switch	out (hardware) + in	interrupts the laser supply; sense contact on a GPIO of the galvo node (F-VLT-30)	student team
Arming button	—	removed 2026-11-10 (ADR-VLT-013): arming is done by software; the kill switch is the only physical control	—
Monitoring laptop	out (from nodes)	push-only monitoring messages, observe-only (ADR-VLT-003)	student team

4.2. Out of scope

- Manual calibration procedures — replaced by automatic calibration (ADR-VLT-012, superseding ADR-VLT-004); only the marker map is measured by hand once, when the cube frame is built.
- Storing camera images by default — GDPR, not needed for the demo (NF-VLT-11).
- Long-term operation, OTA updates, flash-wear management — the project ends after one presentation.
- Certification (CE, EMC, electrical safety, functional safety / SIL) — one-off lab demo, not placed on the market; off-the-shelf boards and supplied power supplies used as delivered.
- Any control path from the monitoring laptop, any network control of the running loop, including over SSH — ADR-VLT-003, ADR-VLT-014, F-VLT-35. SSH for administration is in scope (NF-VLT-10).
- Redundancy of the galvo node and of camera nodes beyond "two of three" — failures lead to the safe state.
- Redundant control node (F-VLT-14) — deferred by the sponsor 2026-11-10 (OQ-8): conflicts with re-arming after the target timeout and needs a membership protocol and a second PTP grandmaster.
- External hardware watchdog for the laser (NF-VLT-4 dropped at the PM review) — replaced by the Pi's on-chip watchdog on the galvo node (NF-VLT-30, ADR-VLT-010). Residual risk: a hang shorter than the watchdog interval leaves the beam frozen at its last clamped aim inside the cube (Class ≤ 2) until the reset or the kill switch (NF-VLT-3).
- Recalibration hint (F-VLT-36, dropped at the PM review) — replaced by the arming check, which starts a galvo scan by itself (F-VLT-31).
- Hardware camera trigger — clock-based fusion instead (ADR-VLT-002).
- Local status indicators on the nodes (LED, buzzer, display) — the monitoring laptop is required to operate the demo and is the only status display (F-VLT-40); safety does not depend on it (NF-VLT-7).
- Arming button and any physical operator control other than the kill switch — arming, disarming, galvo scan and arming check are decided by the control node software (ADR-VLT-013).
- Tracking more than one ball, or objects other than a ball on a string.

5. Decisions taken

Each decision with its reason and who took it. Decisions of the sponsor and given hardware are given to the team, so they are stated as requirements ([L] p. 287: given → requirement). Decisions marked "team, at Gate 1" were taken by the team and approved together with the requirements. Further decisions on the physical layout, the start-up budgets, the camera profile, the timing budget and the error budget are proposed for the architecture (milestone M3); they are not part of this report and are decided at Gate 2.

ID	Decision	Decided by, date	Reason
ADR-VLT-001	Five Raspberry Pi 3 nodes on one wired Ethernet switch	sponsor, 2026-11-02	Separating galvo output from control keeps the safety-relevant actuator logic (clamp, timeout, laser switching) on a node with a single small job. Wired Ethernet gives bounded, predictable latency compared with Wi-Fi.
ADR-VLT-002	One shared clock by software PTP (kernel)	sponsor, 2026-11-02	No extra wiring; works with any camera module; the Pi 3 Ethernet (USB-attached LAN chip) has no hardware timestamping, but software PTP on a quiet

ID	Decision	Decided by, date	Reason
			dedicated switch is typically well below 1 ms (estimate).
ADR-VLT-003	Monitoring on a separate, observe-only laptop	sponsor, 2026-11-02	Keeps display load off the control node; keeps the laptop out of the safety design; lets the team analyse data without touching the loop.
ADR-VLT-004	Manual calibration in v1	superseded by ADR-VLT-012	Manual calibration does not depend on spot visibility, which is itself an open risk (OQ-1, OQ-2, OQ-7).
ADR-VLT-005	Hit = beam hits the ball; aiming error = distance of the beam from the ball centre	sponsor, 2026-11-10	Matches the demo's meaning ("the laser is on the ball"), is measurable with one camera, and gives a meaningful mm value on and off the ball.
ADR-VLT-006	Hit rate is scored by time; gaps count as misses	sponsor, 2026-11-10	Equals the sponsor's original criterion and makes gaps visible instead of removing them.
ADR-VLT-007	One recovery policy: safety faults need re-arming, transient ball loss does not	sponsor, 2026-11-10	Safety-relevant shutdowns need a deliberate decision to resume; short occlusions by the string or the hand must not stop the demo.
ADR-VLT-008	Tracking deadline is firm (p99 30 ms), laser-off deadlines are hard	sponsor, 2026-11-10	A late aim is useless but not dangerous; a late shutdown is the safety-relevant case.
ADR-VLT-009	Time-triggered fusion; galvo aims at the placement time	sponsor, 2026-11-10	A fixed period makes scoring, load and phase analysable; aiming at the placement time removes jitter in transmission from the aim.
ADR-VLT-010	The galvo node uses the Pi's on-chip watchdog	sponsor, 2026-11-10	No additional hardware; turns a hang into a reset, which is already fail-safe.
ADR-VLT-011	Generic constant-velocity / constant-acceleration prediction	sponsor, 2026-11-10	General motion keeps the tracker valid for a hand-moved ball; for the short prediction horizon a constant-acceleration model is close to a pendulum model.
ADR-VLT-012	Automatic calibration of cameras and galvo	sponsor, 2026-11-10	Repeated setup tests need calibration in seconds, not a manual session. The spot on a flat surface is far easier to see than on the ball, which removes ADR-VLT-004's reason for deferring automatic galvo calibration, provided the background allows it (OQ-7).
ADR-VLT-013	Software control by the control node; the kill switch is the only physical control	sponsor, 2026-11-10	The sponsor wants a single physical control. The kill switch already cuts the laser in hardware and is sensed by software, so it can serve as the operator's stop and as the deliberate reset.
ADR-VLT-014	Admin access over SSH, separate from control of the loop	sponsor, 2026-11-10	The team needs to administer five Pis without monitors or keyboards. The safety property that matters is that no network message can influence the loop's decisions. Forbidding all remote access doesn't add to that, and it would block development.
ADR-VLT-015	Given galvo, DAC and laser hardware; converter link faults detected by effect monitoring	given hardware; link-fault policy by the team, 2026-11-10	The hardware is given to the team and could not be chosen, so it is a requirement ([L] p. 287); this record only documents its facts. The team's own decision is how converter link faults are detected: the DAC has no read-back, and a read-back ADC would add hardware and code. The effect-monitoring policy covers qualifying observable deviations and observation loss; it does not establish detection of every electrical link fault. The given laser class and the cube limits remain unchanged.
ADR-VLT-018	Explicit shutdown origins and qualification intervals	team, at Gate 1	Separating qualification, response and watchdog fallback gives each existing limit a clear scope. It avoids strengthening the accepted effect-monitoring hardware decision into a detection guarantee that the chosen interface cannot establish.

ID	Decision	Decided by, date	Reason
ADR-VLT-019	Bounded emission without a measurable optical effect	team, at Gate 1	A time-based policy provides a finite bound without inventing a camera model or a number of missed frames. Distinguishing normal calibration-attempt failure from a Tracking alarm preserves the existing automatic scan/retry behavior.

5.1. Decisions recorded directly as requirements

Question	Decision	Requirement
Operator feedback without the laptop	The laptop is required to operate; no local indicators; on connect it shows current state and recent diagnostics, flags stale nodes; everything is also logged locally	F-VLT-40 , F-VLT-29
Coordinate conventions	Cube frame (marked floor corner, right-handed, mm, z up) and rest position defined in terms; image coordinates of annotations = detection; the rest goes to the architecture	terms, NF-VLT-21
Success criterion and test window	80 % hit rate from release to amplitude < 50 mm	NF-VLT-1, F-VLT-7
Laser class	≤ Class 2	NF-VLT-2
Camera failure behaviour	Degraded mode with two cameras	F-VLT-11
End-to-end deadline	30 ms (sponsor)	NF-VLT-18
Continuous operation	10 min (sponsor)	NF-VLT-12
Numeric defaults (laser timeout 100 ms, monitoring ≤ 20 Mbit/s, CPU ≤ 75 % / RAM ≤ 50 %, run start > 100 mm)	Accepted by the sponsor	F-VLT-7, F-VLT-10, NF-VLT-8, NF-VLT-19
How to verify detection and triangulation without hand-placed checks every time	Recorded, annotated reference data set; automated tests against it (team review)	NF-VLT-21, F-VLT-1 … F-VLT-4, F-VLT-6, F-VLT-7, NF-VLT-17
PM review: hardware watchdog, recalibration hint, redundant control node	NF-VLT-4 and F-VLT-36 dropped; F-VLT-14 first made optional, then deferred on 2026-11-10 (OQ-8)	out of scope, F-VLT-14
Who is responsible for laser safety in the lab?	The student team	NF-VLT-2
Build the redundant control node?	No — deferred	F-VLT-14, out of scope
80 % also with two cameras?	Yes, for every camera pair	NF-VLT-1

6. Open questions

No question about what the system must do is open: every requirement states the rule that has to hold. Open are numeric values the team chooses and facts of the given galvo measured on the bench. Chosen values belong in the architecture ([L] p. 287), so they are deferred there. Until they are fixed, the requirements listed under “Needed by” can be tested for their rule, but not yet at their limit. Owners are node teams.

6.1. Deferred to the architecture

These questions concern values the team chooses, so they belong in the architecture ([L] p. 287), or (OQ-4) facts of the given galvo that the design needs and that are measured on the bench. Each has a proposal, which the team decides in the architecture consistency review of milestone M3.

ID	Question	Proposal	Owner	Needed by
OQ-3	Cube size and clamp envelope (the starting deflection is fixed in NF-VLT-1).	Clear cube 800 × 800 × 800 mm, nominal rest centre (400, 400, 300) mm, clamp envelope 0 … 800 mm per axis.	Camera team, laser team	F-VLT-9 envelope values, the cube configuration
OQ-4	Facts of the galvo set (ADR-VLT-015) needed for the design: driver input wiring (single-ended or differential)	The output chain galvo node → PCM5102A → driver → mirrors already works without a pattern	Laser team	the galvo node design, the settling

ID	Question	Proposal	Owner	Needed by
	and connector, usable optical scan angle after level matching (the DAC alone gives ≈ ±8.9°; a draft layout needs ≈ ±14.2°, NF-VLT-32), angle per volt and step response of the delivered unit.	board, with the supplied power supply; the driver's trim potentiometers set the gain for the full range; wiring type, usable angle with the ≥ 10 % margin of NF-VLT-32, angle per volt and step response are measured on the bench.		budget of NF-VLT-18
OQ-7	Background and ball colour (the laser is red, so the ball must not be red); the spot must be visible on floor and walls for the galvo scan (F-VLT-13) and the arming check (F-VLT-31), and the frame markers (F-VLT-12) need contrast.	Matte white ball, matte medium-grey floor and far walls, ordinary room lighting, exposure and white balance set at start and then locked.	Camera team	NF-VLT-17, F-VLT-12/13/31
OQ-12	Deadline classes are resolved (ADR-VLT-008). Still open: maximum observation age, exposure-time skew, prediction horizon, target age, sync-status age, fusion period, global granularity g (NF-VLT-24), and the conversion of the camera's exposure timestamp to the shared clock, including rolling-shutter row timing.	Observation age ≤ 25 ms, skew ≤ 20 ms, prediction horizon ≤ 30 ms, target age ≤ 5 ms, sync-status age ≤ 100 ms, fusion period 10 ms, g = 2 ms, timestamp mapping error ≤ 0.5 ms; to be checked against the modes of the given camera.	Control team, camera team	F- , VLT-3/4/20/22/23 NF-VLT-24/27
OQ-14	The degraded-mode part (every camera pair must reach 80 %) and the operating envelope (decided 2026-11-12) are resolved in NF-VLT-1. Still open: one combined error budget for detection, calibration, triangulation, timing/ prediction and galvo placement against the ball radius.	Perception ≤ 10 mm, prediction and timing ≤ 8 mm, galvo placement ≤ 5 mm (all p99), reserve 2 mm against the 25 mm ball radius.	Camera team, control team, laser team	the error budget of the architecture; it may tighten the limits of F-VLT-3/4 by a change request

6.2. Known risks

Risks that do not change a requirement now, but may need a change request after Gate 1.

ID	Risk	Owner	Affects
R-1	The given camera may not meet the tracking deadline. Its high-rate 640 × 480 mode (≈ 103 fps) uses a cropped sensor area, so the field of view shrinks to ≈ 26.5° × 20°, below the ≥ 45° that the proposed layout needs. The full-view 1640 × 1232 mode keeps 62.2° × 48.8° but reaches only ≈ 42 fps (≈ 24 ms frame period), before readout, rolling-shutter correction and detection on a Pi 3. NF-VLT-18 (p99 ≤ 30 ms, given by the sponsor) stays unchanged. Check: bench measurement of the camera mode, readout and detection time in M3, before Gate 2. If it fails: a change request to NF-VLT-18 or to the layout (camera distance), decided by the team with the sponsor.	Camera team	NF-VLT-18, NF-VLT-1, NF-VLT-32

6.3. Resolved questions

- OQ-1 ⟶ Raspberry Pi Camera Module v2.1 (Sony IMX219, rolling shutter) on Raspberry Pi 3 Model B+, given hardware (given components)
- OQ-2 ⟶ ADR-VLT-015 (Laserfuchs LFD650-1-12, red, 1 mW, Class 2)
- OQ-5 ⟶ ADR-VLT-015 (PCM5102A I2S DAC; link faults detected by effect monitoring)
- OQ-6 ⟶ NF-VLT-2
- OQ-8 ⟶ F-VLT-14 deferred (out of scope)
- OQ-9 ⟶ ADR-VLT-005
- OQ-10 ⟶ ADR-VLT-006
- OQ-11 ⟶ ADR-VLT-007
- OQ-13 ⟶ calibration procedure ⟶ ADR-VLT-012; cube frame and rest position defined in terms
- OQ-15 ⟶ ADR-VLT-009
- OQ-16 ⟶ ADR-VLT-010

- OQ-17 ⟶ ADR-VLT-011

7. Gate 1

The team has read and approved the requirements in this report (F-VLT-1 … F-VLT-42, NF-VLT-1 … NF-VLT-34) and the decisions in Section 5 on 12 November 2026. The architecture (milestone M3) starts after this approval.